No of Pages :4 Course Code: 23MX14

Roll No:
(To be filled in by the candidate)

PSG COLLEGE OF TECHNOLOGY, COIMBATORE 641 004

SEMESTER EXAMINATIONS, DECEMBER 2023

MCA Semester: 1

23MX14 DATABASE SYSTEMS

Time : 3 Hours **Maximum Marks: 100**

INSTRUCTIONS:

1. Answer **ALL** questions. Each question carries 25 Marks.
2. Subdivision (a) carries 3 marks each, subdivision (b) carries 10 marks each and subdivision (c) carries 12 marks each.
3. Course Outcome : Table

Qn.1	CO1	Qn.2	CO2	Qn.3	CO3	Qn.4	CO4

1. a) Define composite, multivalued and derived attributes. Give examples for each?

b) i) With a neat block diagram, outline how data is accessed in a DBMS using its various components. [5]

ii) Consider the following applications. You work for a large online shopping company, and maintain their orders database. They have tens of millions of orders, and need to access about 2-300 orders per second; orders are retrieved either by the order ID, or by the date, or by the customer. What data model would you use for the orders database? Is it better to use a file processing system or a database for storage? What problems do you foresee? Justify. [5]

c) Consider a database system for a baseball organization. The personnel involved in the league include players, coaches, managers, and umpires. Each is identified by a unique personnel id. They are also described by their name along with the date and place of birth. Players are further described by other attributes such as their batting orientation (left, right, or switch) and have a lifetime batting average (BA).

Within the players group is a subset of players called pitchers. Pitchers have a life time ERA (earned run average) associated with them.

Teams are uniquely identified by their names. Teams are also described by the city in which they are located and the division and league in which they play (such as Central division of the American league). Teams have one manager, a number of coaches, and a number of players.

Games are played between two teams with one designated as the home team and the other the visiting team on a particular date. The score (runs, hits, and errors) are recorded for each team. The team with more number of runs is declared the winner of the game.

With each finished game, a winning pitcher and a losing pitcher are recorded. In case there is a save awarded, the save pitcher is also recorded.

With each finished game, the number of hits (singles, doubles, triples, and home runs) obtained by each player is also recorded. Design an Entity-Relationship diagram for the baseball database based on above requirements by specifying relationships types, connectivity, cardinality and key attributes. Also convert into relational schema.

2. a) Distinguish between independent and correlated nested queries. Provide appropriate examples to support your explanation.

   b) Explain the significance of enforcing integrity constraints in a relational data model with suitable examples. In the given two relations, A is the primary key of R and D is the primary key of S and there is a referential Integrity between S.A and R.A, List all integrity constraints that are violated.

R

A	B	C
A1	b1	c1
Null	b2	null
a1	b1	c1

S

D	A	F
d1	a1	f1
d1	a2	null

   c) Explain the significant use of relational algebraic expressions in querying data from database with examples. Consider a relational database about hotels, customers (guests) and their bookings that are maintained by an online hotel booking company. The database consists of the following relations.

   Hotel (hId, hName, hAddress, hCity)
   Guest (gId, gName, gAddress, gCity)
   Room (hId, roomNo, type, price)
   Booking (gId, hId, roomNo, fromDate, year, noOfDays)
   Set necessary constraints for the above relation. Write relational algebraic expressions for the following queries.

   i) List the names of hotels located in the city "Coimbatore"

   jj) List the names of guests who have stayed in the hotel "Residency" and in the room number 100.

   iii) List the ids of guest who have booked at least one room of type "suite" in every hotel located in "Chennai".

   iv) Find the details of hotel with room rent (price) value greater than all the hotels in that city.

   v) Find the details of the hotel which is located in at least one location in the city "Coimbatore" and which does not exists in "Chennai".

3. a) What are the significant use of rowid and rownum in Oracle database? How can we extract the nth record and top-k records from the table?

   b) i) Explain how "group by" clause works. What is the difference between the WHERE and HAVING clauses? [5]

   ii) The following two relational algebra expressions e1 & e2 are equivalent that is, they return the same answer when evaluated on any instance of the database. Justify your answer with reasoning. [5]

   e1) $\pi_{title,year}(\sigma_{length \geq 100}(Movie) \cap \sigma_{studio='Fox'}(Movie))$

   e2) $\pi_{title,year}(\sigma_{length \geq 100 \wedge studio='Fox'}(Movie))$

c) i) Explain the various methods to implement Join operations using SQL statements. Consider the following two relations Student and Employee. Write relational algebra statements and equivalent SQL to perform inner join and all Consider the STUDENT and EMPLOYEE relations, Express the following queries in relational algebra.

STUDENT		
ROLL	NAME	AGE
1	AMAR	20
2	AKIL	18
3	BALA	19
4	HARSHA	20
5	PRADEEP	21
6	PRADEEP	23

EMPLOYEE		
EMPLOYEE_NO	NAME	AGE
E1	ANAND	20
E2	AKASH	23
E3	BALA	25
E4	HARSHA	20
E5	PRANAV	22

1) Find the students with "AGE" 20.

2) Find the names which are in STUDENT relation but not in EMPLOYEE relation.

3) Find the output of the query π_{NAME}(STUDENT) Ω π_{NAME}(EMPLOYEE).

4) Find the names of the employees whose age is > the maximum age of student.

5) Find the names of student and employees whose name starts with 'A'.

(OR)

ii) Illustrate the following concepts with examples

1) View scan be used to protect data access

2) Indexing improves the performance of query processing

3) Triggers

4) Discuss **two** scenarios where a view can be helpful in a database management system. Explain why it is not always possible to perform SQL UPDATE /DELETE/ INSERT statements on top of a view.

4. a) What are the significance of the commit, rollback and save point in SQL for the transaction management?

b) i) Detail the various data base security measures and privacy mechanisms to be provided during database design. How the GRANT and REVOKE commands can be used for access control? [5]

ii) Consider a relation R with six attributes A, B, C, D, E and F. Given a set of functional dependencies B→E,F for the relation schema R(A, B, C, D, E, F) with primary key AB under which R is in 2NF but not in 3NF. Justify your answer. [5]

c) i) Explain the purpose of normalization process in database design life cycle. The project management report is given below:

Project Management Report

Project Code: PC010
Project Title: Pensions System

Project Manager: M Phillips **Project Budget:** £24,500

Employee No.	Employee Name	Department No.	Department Name	Hourly Rate
S10001	A Smith	L004	IT	£22.00
S10030	L Jones	L023	Pensions	£18.50
S21010	P Lewis	L004	IT	£21.00
S00232	R Smith	L003	Programming	£26.00

Total Staff on Project: 4 **Average Hourly Rate:** £21.88

Calculated Fields

Identify the functional dependency exists and convert this form into a relational model such that it satisfies the rule of 3rd normal form. Does the result of your decomposition preserve dependencies?

(OR)

ii) Explain the significance of ACID properties of a transaction. How to preserve the isolation property and concurrency of the concurrently executing transactions? Explain the two-phase locking protocol in detail.

Consider a database consisting of a single relation R:

R:

A	B
1	10
2	0

The following two transactions run concurrently on this database:

Line	T1	T2
1	begin transaction;	begin transaction;
2	update R set B = B-10 where A=1;	select sum(B) from R;
3	update R set B = B+10 where A=2;	commit;
4	commit;	

Is it ever possible for T2 to see a value of zero in its output? Explain why or why not.

/END/

FD/RL